

Homework 2: Limits and Open Sets

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Due: Chapter 2, ETC: 1-2 hours, 100 points

Practice Problems: Work together in class

1. For each sequence below, decide whether it converges. If it converges, find its limit.

(a) $x_n = \frac{1}{n}$

(b) $x_n = 5 + \frac{2}{n}$

(c) $x_n = n$

(d) $x_n = (-1)^n$

2. Suppose the price of a good in period n is given by

$$p_n = 20 + \frac{10}{n}.$$

(a) Compute p_1 , p_2 , and p_{10} .

(b) Does the price sequence converge?

(c) If it converges, what is the limiting price?

(d) Give a brief economic interpretation of the limit.

3. Suppose a firm's output in period n is

$$q_n = 100 - \frac{20}{n}.$$

(a) Compute q_1 , q_2 , and q_{20} .

(b) Does $\{q_n\}_{n=1}^{\infty}$ converge?

(c) If it converges, find $\lim_{n \rightarrow \infty} q_n$.

(d) Is the sequence increasing, decreasing, or neither?

4. Consider two consumption bundles

$$\mathbf{x} = (2, 3), \quad \mathbf{y} = (5, 7).$$

Using the Euclidean metric,

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2},$$

find the distance between $\mathbf{x}$ and $\mathbf{y}$.

5. Let

$$B_2((1, 1)) = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x} - (1, 1)\| < 2\}.$$

For each point below, determine whether it belongs to $B_2((1, 1))$.

- (a) $(1, 1)$
- (b) $(2, 1)$
- (c) $(3, 1)$
- (d) $(2, 2)$

Homework Problems: Submit individually

6. For each sequence below, decide whether it converges. If it converges, find its limit.

- (a) $x_n = \frac{3n+1}{n}$
- (b) $x_n = \frac{2}{n^2}$
- (c) $x_n = 4 - \frac{1}{n}$
- (d) $x_n = 2 + (-1)^n$

7. Suppose a consumer's planned savings in period n are given by

$$s_n = 500 + \frac{100}{n}.$$

- (a) Find s_1 , s_5 , and s_{100} .
- (b) Does the sequence converge?
- (c) If it converges, find its limit.
- (d) Interpret the limiting value as a long-run planned savings level.

8. Let

$$S = (0, 10) = \{x \in \mathbb{R} : 0 < x < 10\}.$$

Interpret S as the set of feasible prices for a good.

- (a) Is S open?
- (b) Is S closed?
- (c) What are the boundary points of S ?
- (d) Is the price $p = 0$ in S ?
- (e) Is the price $p = 10$ in S ?

9. Let

$$T = [0, 10] = \{x \in \mathbb{R} : 0 \leq x \leq 10\}.$$

- (a) Is T open?
- (b) Is T closed?
- (c) What are the boundary points of T ?
- (d) Give an example of a sequence contained in T that converges to 10.
- (e) Does the limit of your sequence belong to T ?

10. Let the feasible set of consumption bundles be

$$X = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0, x_2 > 0\}.$$

- (a) Is the bundle $(1, 1)$ in X ?
- (b) Is the bundle $(0, 1)$ in X ?
- (c) Is X open in $\mathbb{R}^2$?
- (d) Is X closed in $\mathbb{R}^2$?
- (e) Consider the sequence $\mathbf{x}_n = (1/n, 1)$. Is every term of this sequence in X ?
- (f) What is the limit of $\mathbf{x}_n$?
- (g) Does the limit belong to X ? What does this tell you about whether X is closed?